ESSENTIALLY CONSTANT INVERSE SYSTEMS AND THE MIDDLE TRIANGLE ENTRY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 12659-12704, following the statement at 12632-12657.

Editorial note. No omissions: the 5-lemma step displays the two long exact rows; the independence of delta from n follows from the identity transitions on the split-off summands.

Allowed standard prerequisites (editorial).

- Essentially constant systems split off the limit value. lemma-essentially-constant (same chapter, cited at ~12644). Lets one renumber so that A_n = A + A'_n and C_n = C + C'_n with essentially zero tails.
- Pro-category recognition. Categories, Remark categories-remark-pro-category-copresheaves and categories-remark-pro-category (cross-chapter, cited at the end ~12699-12703). Converts the Hom-isomorphism statement into essential constancy of (B_n).

Source reference. Original reference 05PX: focused-seeds/geometry/S9-der-essentially-constant-2-out-of-3/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G2X: focused-seeds/geometry/S9-der-essentially-constant-2-out-of-3/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0G39: sources/stacks-derived/pinned-original-source.tex, source lines 12605–12605. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let*

$$A_n \to B_n \to C_n \to A_n[1]$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

be an inverse system of distinguished triangles in $\mathcal{D}$. If (A_n) and (C_n) are essentially constant, then (B_n) is essentially constant and their values fit into a distinguished triangle $A \to B \to C \to A[1]$ such that for some $n \geq 1$ there is a map

$$\begin{array}{ccccccc} A_n & \longrightarrow & B_n & \longrightarrow & C_n & \longrightarrow & A_n[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & A[1] \end{array}$$

of distinguished triangles which induces an isomorphism $\lim_{n' \geq n} A_{n'} \to A$ and similarly for B and C.

Proof. After renumbering we may assume that $A_n = A \oplus A'_n$ and $C_n = C \oplus C'_n$ for inverse systems (A'_n) and (C'_n) which are essentially zero, see Lemma 0G39. In particular, the morphism

$$C \oplus C'_n \to (A \oplus A'_n)[1]$$

maps the summand C into the summand $A[1]$ for all n by a map $\delta : C \to A[1]$ which is independent of n. Choose a distinguished triangle

$$A \to B \to C \xrightarrow{\delta} A[1]$$

Next, choose a morphism of distinguished triangles

$$(A_1 \to B_1 \to C_1 \to A_1[1]) \to (A \to B \to C \to A[1])$$

which is possible by TR3. For any object D of $\mathcal{D}$ this induces a commutative diagram

$$\begin{array}{ccccccccc} \ldots & \longrightarrow & \mathrm{Hom}_{\mathcal{D}}(C, D) & \longrightarrow & \mathrm{Hom}_{\mathcal{D}}(B, D) & \longrightarrow & \mathrm{Hom}_{\mathcal{D}}(A, D) & \longrightarrow & \ldots \\ & & \downarrow & & \downarrow & & \downarrow & & \\ \ldots & \longrightarrow & \mathrm{colim}\, \mathrm{Hom}_{\mathcal{D}}(C_n, D) & \longrightarrow & \mathrm{colim}\, \mathrm{Hom}_{\mathcal{D}}(B_n, D) & \longrightarrow & \mathrm{colim}\, \mathrm{Hom}_{\mathcal{D}}(A_n, D) & \longrightarrow & \ldots \end{array}$$

The left and right vertical arrows are isomorphisms and so are the ones to the left and right of those. Thus by the 5-lemma we conclude that the middle arrow is an isomorphism. It follows that (B_n) is isomorphic to the constant inverse system with value B by the discussion in Categories, Remark 0G2X. Since this is equivalent to (B_n) being essentially constant with value B by Categories, Remark 05PX the proof is complete. □